Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.7 Plans and Elevations

Plans and Elevations

STARTER · 5 MIN

Date: _____ **Title: Plans and Elevations**

LEARNING OBJECTIVES

- Use 2D representations of 3D solids

STARTER *Answer in your book. No calculators.*

1. Faces of a square-based pyramid?
2. Faces of a triangular prism?
3. Faces of a pentagonal prism?
4. Faces of a hexagonal pyramid?

Starter — Answers

STARTER

Tick or correct in green pen

1. 1 square, 4 triangles
2. 2 triangles, 3 rectangles
3. 2 pentagons, 5 rectangles
4. 1 hexagon, 6 triangles

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify the shapes of the faces of 3D solids	☐
SILVER	I can draw the plan, front and side elevations of a solid	☐
GOLD	I can draw 3D solids on isometric paper	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Plans and Elevations

INTRODUCTION · 3 MIN

A 3D solid can be drawn in 2D as three views.

Plan

The view from above.

Looking down on the solid

Front elevation

The view from the front.

Looking straight at the front

Side elevation

The view from the side.

Looking from the right

Isometric paper

Draw 3D solids on dotted paper.

Vertical lines stay vertical

Examples 1 and 2 — Faces of Solids

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Prism: 2 ends + rectangles Pyramid: base + triangles

Example 1

What shapes are the faces of a square-based pyramid?

Example 2

What shapes are the faces of a pentagonal prism?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

A base and triangles that meet at the top

Base = 1 square

Sides = 4 triangles

Tip: one triangle on each side of the base.

Example 2

Two ends and rectangles

Ends = 2 pentagons

Sides = 5 rectangles

Tip: one rectangle for each side of the end.

Task 1

PRACTICE · 9 MIN

BRONZE

Name the shapes of the faces

Hint Prism: 2 ends + rectangles Pyramid: base + triangles

1. Triangular prism
2. Cube
3. Hexagonal pyramid
4. Cuboid
5. Triangular-based pyramid

★ Hexagonal prism

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 2 triangles, 3 rectangles
2. 6 squares
3. 1 hexagon, 6 triangles
4. 6 rectangles
5. 4 triangles

★ 2 hexagons, 6 rectangles

Examples 3 and 4 — Plan, Front and Side Elevations

EXAMPLES · 5 MIN

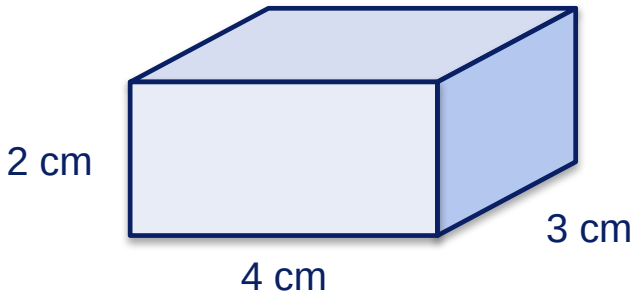
SILVER

Watch and copy into your book

Hint Plan: from above Front, side: straight on

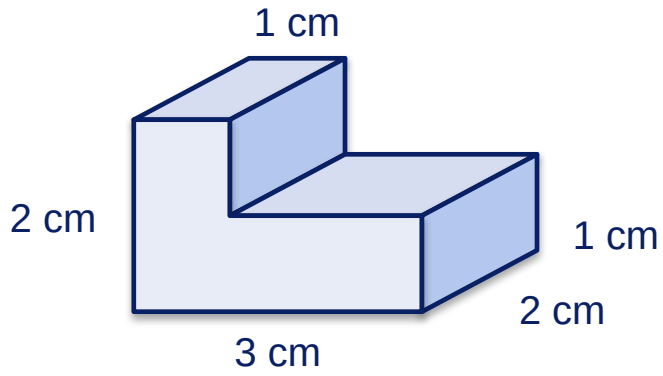
Example 3

Draw the plan, front elevation and side elevation.



Example 4

Draw the plan, front elevation and side elevation.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Three rectangles

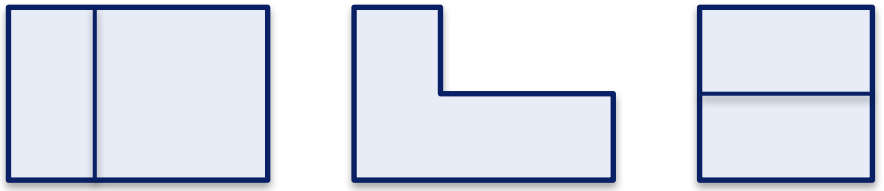


Plan Front Side

Tip: plan 4 by 3, front 4 by 2, side 3 by 2.

Example 4

Draw the edges you can see



Plan Front Side

Tip: the step shows as a line.

Task 2

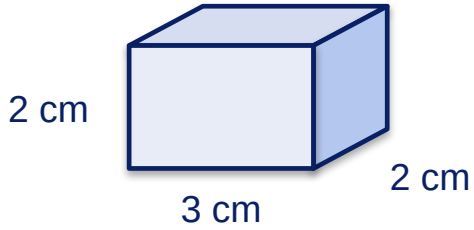
PRACTICE · 9 MIN

SILVER

Draw the plan, front and side elevations

Hint Plan: from above Front, side: straight on

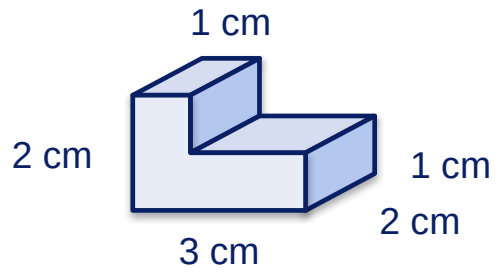
1 Plan, front, side



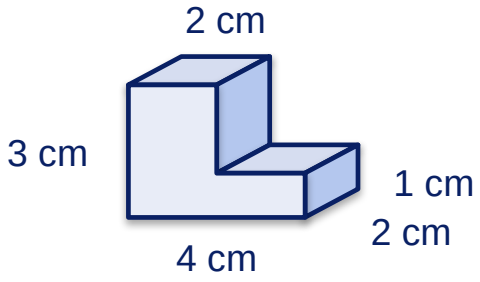
2 Plan, front, side



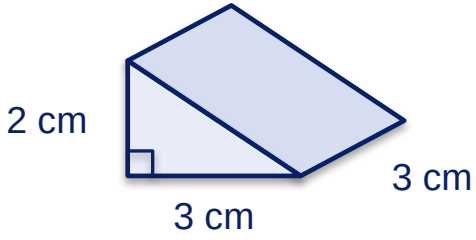
3 Plan, front, side



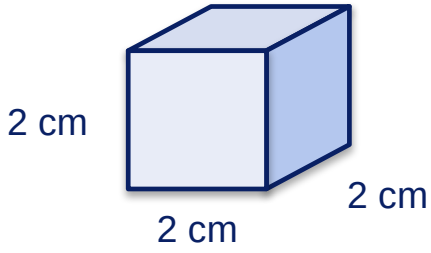
4 Plan, front, side



5 Plan, front, side



★ Plan, front, side



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1



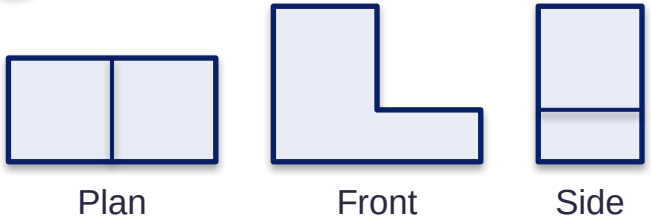
2



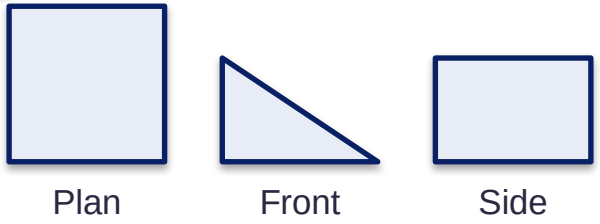
3



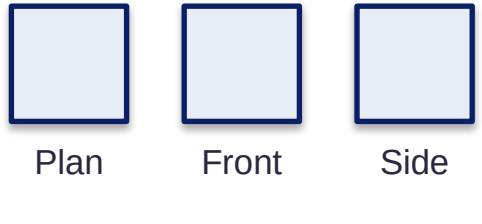
4



5



★



Examples 5 and 6 — From Views to Solids

EXAMPLES · 5 MIN

GOLD

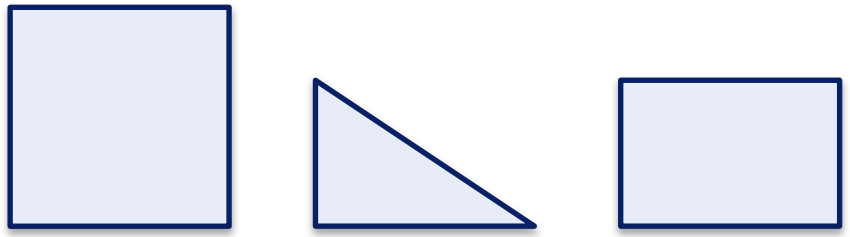
Exam-style: show all your working

Hint Look at the plan first Match each view

Example 5

(1 mark)

Name the solid.

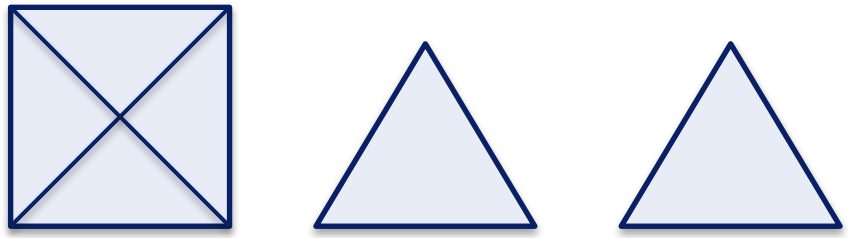


Plan Front Side

Example 6

(1 mark)

Name the solid.



Plan Front Side

Examples 5 and 6 — Corrections

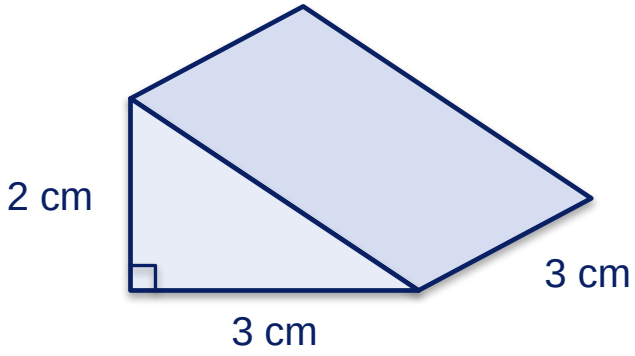
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

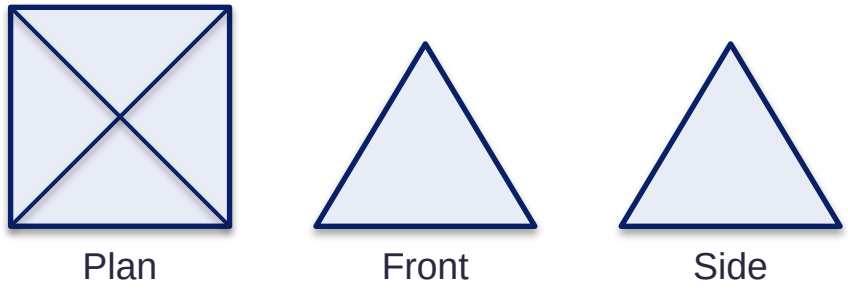
Triangle front, rectangles elsewhere



Tip: it is a triangular prism.

Example 6

Square plan, triangle front and side



Tip: it is a square-based pyramid.

Task 3

PRACTICE · 9 MIN

GOLD *Name the solid*

Hint Look at the plan first Match each view

1. Plan: square
   Front and side: triangles

2. Plan: circle
   Front and side: rectangles

3. Plan: circle
   Front and side: triangles

4. Plan, front and side:
   equal squares

5. Front: triangle
   Plan and side: rectangles

★ Plan, front and side:
   equal circles

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **Square-based pyramid**
2. **Cylinder**
3. **Cone**
4. **Cube**
5. **Triangular prism**
★ **Sphere**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE What does the plan show?

SILVER Front elevation of a cuboid 5 cm wide and 3 cm high?

GOLD Plan: square. Front and side: triangles. Name the solid.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **The view from above**

SILVER **A 5 cm by 3 cm rectangle**

GOLD **Square-based pyramid**

Exam Practice — Faces and Views

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
Prism: 2 ends + rectangles
Pyramid: base + triangles

Question 1

(2 marks)

Write down the shapes of the faces of a triangular prism.

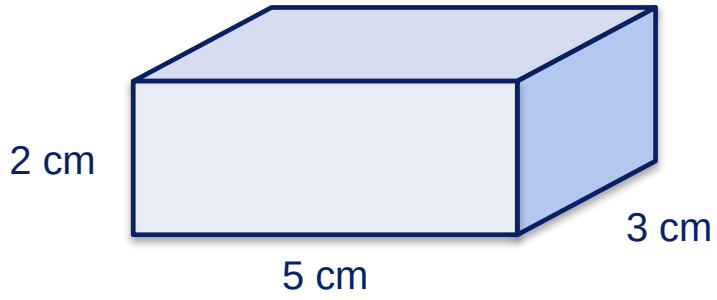
.....

Question 2

(2 marks)

Here is a cuboid.

Diagram NOT accurately drawn



(a) Sketch the plan. (1)

.....

(b) Sketch the front elevation. (1)

.....

Exam Practice — Elevations

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

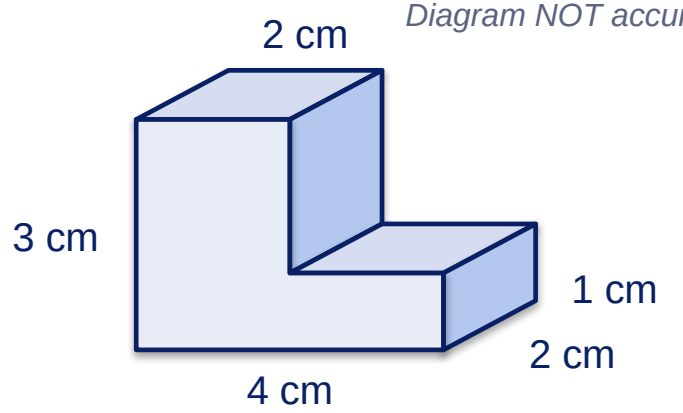
Hint Plan: from above Front, side: straight on

Question 3

(3 marks)

Draw the plan, front elevation and side elevation.

Diagram NOT accurately drawn



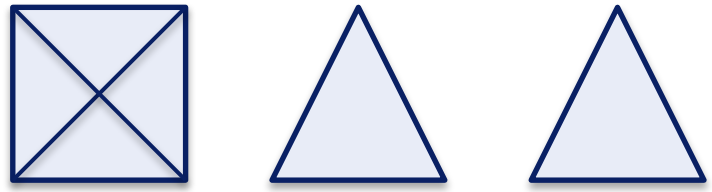
.....

Question 4

(2 marks)

Here are three views of a solid.
Name the solid.

Diagram NOT accurately drawn



.....

Exam Practice — Views and Solids

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

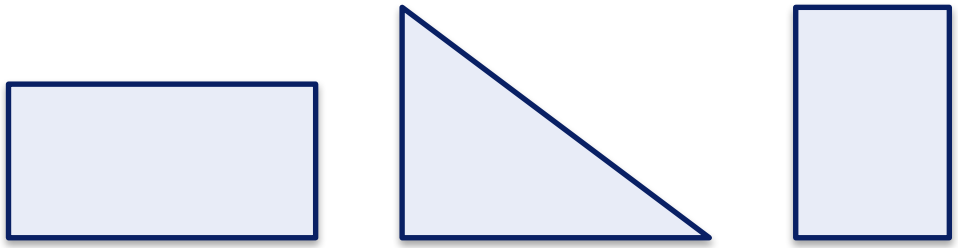
Hint Look at the plan first Match each view

Question 5

(2 marks)

Here are three views of a solid.
Name the solid.

Diagram NOT accurately drawn



.....

Question 6

(3 marks)

The plan of a solid is a circle.
Its front and side elevations are triangles.

(a) Name the solid. (1)

.....

(b) How many faces? (1)

.....

(c) How many vertices? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

2 triangles B1

3 rectangles B1

Tip: two ends and three sides.

Question 2

(a) rectangle 5 cm by 3 cm B1

(b) rectangle 5 cm by 2 cm B1

Tip: plan is from above.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

plan: 4 by 2, line at 2 from the left — B1

front: L-shape, step 1 cm high — B1

side: 2 by 3, line at 1 cm — B1

Tip: show the step as a line.

Question 4

square-based pyramid — B2

B1 for pyramid

Tip: the plan shows the square base.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

triangular prism B2

B1 for prism

Tip: the front shows the end.

Question 6

(a) cone B1

(b) 2 (1 flat, 1 curved) B1

(c) 1 B1

Tip: the apex is the vertex.